Hazem Mohsen Abdelaziz Mohamed

Robotics and Industrial Automation Engineer

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Professional Summary

Mechatronics Engineer specializing in robotics, autonomous systems, and industrial automation, with experience in engineering education, multi-robot systems, and autonomous vehicle development. Skilled in ROS 2, MATLAB/Simulink, SLAM, sensor fusion, computer vision, and PLC-based control. Interested in robotics R&D, autonomous systems, and practical engineering solutions.

Professional Experience

Teaching Assistant *Aug 2026 – Present*

German University in Cairo (GUC), Faculty of Engineering and Materials Science *Full-time*

- Support undergraduate laboratory teaching in robotics, pneumatics, and programmable logic controllers (PLCs) within the Mechatronics Engineering program.
- Assist with robotics simulation, control-system implementation, and MATLAB/Simulink-based engineering applications.
- Guide students through practical exercises, control logic, automation principles, and troubleshooting.
- Support hands-on learning using industrial automation tools, including Siemens TIA Portal and Factory I/O.

Autonomous vehicles simulation tester *Jun 2026 – Aug 2026*

Micro-Engineering Tech. inc. *Full time*

- Performing Road-to-Simulation (R2S) test cases for GM-USA autonomous vehicle (Cruise)
- Optimizing test scenarios for different autonomous vehicles behavior to be reported to development team.
- Creating and Utilizing automation scripts to generate a robust approach for different test suits

Research Assistant – Multi-Robot Systems Lab *Nov 2025 – Jan 2026*

German University in Cairo (GUC) *Part-time*

- Contribute to cooperative autonomous vehicle and multi-robot systems development.
- Work with ROS 2-based robotic architectures, embedded vehicle control, simulation, and system integration.
- Investigate risk-aware decision-making and collision-risk assessment using Time-to-Collision (TTC).
- Contribute to simulation and hardware validation using Webots and scaled autonomous vehicle platforms.

Projects

Robust Visual SLAM Using MATLAB Mobile Streaming *MathWorks Project Challenge #213*

- Developed a MATLAB App Designer application for visual SLAM and pose estimation using camera and IMU data streamed from a mobile device.
- Explored indoor and outdoor localization workflows, including GPS-supported outdoor positioning.
- Developing a real-time processing pipeline for data aquired fro mobile phone to the desktop application.

Autonomous Vehicle Localization Using Onboard Sensors *MathWorks Project Challenge*

- Worked on sensor-based localization and fusion using LiDAR, camera, IMU, and GPS data.
- Explored point-cloud registration and adaptive sensor-fusion techniques for autonomous vehicle localization.

ROS 2-Based Cooperative Autonomous Scaled Vehicle *Multi-Robot Systems*

- Integrated ROS 2 software with embedded vehicle control using Raspberry Pi and microcontrollers.
- Worked with custom ROS 2 messages, motor and steering control, and Docker-based deployment.

LiDAR-Based Mapping and Localization *ROS 2 / Point Cloud Processing*

- Processed recorded LiDAR datasets for mapping and localization experiments.
- Explored Iterative Closest Point (ICP) registration and comparison against ENU odometry ground truth.

Education

B.Sc. in Mechatronics Engineering *July 2024*
Faculty of Engineering, Ain Shams University
Cumulative GPA: 3.09/4.00

Technical Skills

Robotics: ROS 2, SLAM, localization, sensor fusion, robot kinematics, autonomous systems, multi-robot systems

Programming: C++, Python, MATLAB

Simulation and Engineering: MATLAB/Simulink, Webots, Gazebo, CoppeliaSim, RViz2, OpenCV, Simscape Multibody

Industrial Automation: Siemens TIA Portal, PLC programming, Factory I/O, pneumatics, industrial control

Embedded Systems and Tools: Raspberry Pi, microcontrollers, Linux, Docker, Git

Selected Certifications

- **State Estimation and Localization for Self-Driving Cars** – MathWorks
- **Robust VSLAM** – MathWorks
- **Code Foundation for ROS** – The Construct
- **ROS Basics: Program Robots!** – Udemy